

Edrill Bilan

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Education

Pamantasan ng Lungsod ng Maynila BS in Computer Science <i>Tertiary Education</i> – Magna Cum Laude, GWA: 1.35 (Graduation: Aug. 2026)	Sept 2022 – Present
Universidad de Manila STEM Strand <i>Senior High School</i> – Graduated with High Honors, GWA: 95.0	Oct 2020 – July 2022

Skills

Programming Languages: Python, Java, C, SQL, JavaScript, TypeScript, HTML, CSS, x86 Assembly
Machine Learning: PyTorch, TensorFlow, ChromaDB, RAG, scikit-learn, Pandas, NumPy, Matplotlib, SciPy
Backend Development: MySQL, PostgreSQL, Django, Flask, ExpressJS, FastAPI, Firebase, Supabase
Frontend & Other Tools: ReactJS, NextJS, Tailwind CSS, JavaFX, Git, Android Studio, Figma, Vercel, Ollama

Experience

Crawlers Insurance Agency, Inc. AI Intern · Remote, Quezon City, Philippines	June 2026 – Present
Developed CrawlLearn, an insurance edtech platform featuring an LLM-based learning assistant, AI-narrated video lessons, and an interactive transcript synced in real time with lesson playback.	
National Cheng Kung University Research Intern · Computer Vision and Multimedia Laboratory, Tainan, Taiwan	Oct 2025 – Nov 2025
International research collaboration with NCKU's Computer Vision and Multimedia Laboratory, finetuning ResNet-50 w/ multi-scale transformers for Pulmonary Lung Nodule Detection on Chest X-ray images as part of undergraduate thesis; published on MDPI's Q1 Journal of Diagnostics.	
Goldilocks Bakeshop, Inc. IT Service Management Intern · Greenfield Building, Mandaluyong City	June 2025 – July 2025
Assisted in maintaining IT infrastructure while coordinating service requests such as technical support for hardware & software issues, streamlining issue resolution workflows across departments.	

Projects

(Machine Learning) OmniJuris – PHL Legal Intelligence RAG System (2026)	
- Developed a RAG pipeline over 3.2M chunks of the Philippine OmniCorpus dataset with multilingual BGE-M3 embeddings, ChromaDB vector search, and cross-encoder reranking - Integrated a locally-hosted Qwen3 4B LLM via Ollama with SQLite-backed conversational memory for context-aware legal queries - Stack: FastAPI, React, TypeScript, ChromaDB, sentence-transformers, Ollama, SQLite	
(Machine Learning) Atlas990 – USA Nonprofit Organizations Intelligence System (2026)	
- Cleaned raw public IRS Form 990 tax data into an optimized Parquet format - Developed a full-stack web application that uses an XGBoost model to calculate a 0–100 priority score for nonprofit leads and integrates Meta's FAISS library to instantly find matching lookalike organizations. - Stack: React, Vite, TypeScript, FastAPI, Pandas, NumPy, scikit-learn, XGBoost, FAISS	
(Deep Learning) Pulmonary Lung Nodule Detection System (2026)	
- Developed, trained, and finetuned a ResNet-50 deep learning model enhanced with light transformer blocks per stage for early lung cancer screening on chest X-rays as part of undergraduate thesis. - Stack: Python, PyTorch, OpenCV, Pandas, NumPy	

(Web Crawler) TrabaHound (2026)

- Developed a comprehensive job aggregation platform featuring a Next.js and TypeScript frontend, serverless APIs, and a Python crawler Jobspy.
- Implemented PDF parsing and LLM-based resume analysis for enhanced job matching.
- Stack: NextJS, TypeScript, Python, Jobspy, LLMs, PDF Parsing

(Full-Stack) MAFA-Inventi (2025)

- Co-developer for a property management system developed for the Inventi Hackathon enabling property managers to upload floor plans, pin issues, and efficiently manage properties, tenants, and maintenance.
- Implemented an LLM-based dashboard analysis.
- Stack: ReactJS, Tailwind CSS, NodeJS, ExpressJS, Supabase, Vercel

(Full-Stack) Bilandog Corporation E-commerce Website (2025)

- A personal project developed for a fictional e-commerce business to improve proficiency in Full-Stack Web Development, practicing RESTful principles and database design.
- Stack: Django, PostgreSQL, Node.js, ReactJS (Next.js), TailwindCSS

(Full-Stack) DOM Compiler (2024)

- Lead developer for the DOM programming language, designed with custom personalized syntax and keywords to meet Compiler Design course requirements.
- Stack: NextJS, Tailwind CSS, Flask, React, NodeJS

Extracurricular

- **Data Engineering Pilipinas | DataCamp Scholar**
DataCamp x Data Engineering Pilipinas Partnership Scholar 2026
- **Inventi Asia Hackathon | Creativity Award Winner**
Smart Property Solutions Hackathon 2025
- **Google Developer Student Club (GDSC) Volunteer | Web Development Team**
October 2024 – October 2025
- **Google Developer Student Club (GDSC) Volunteer | Design and Branding Team**
November 2023 – October 2024
- **DSPC Annual Journalism Competition | 3rd Place Winner in Online Publishing (English)**
Manila Division Schools Press Conference 2019

Certifications

- Intermediate Machine Learning – **Kaggle** (2026)
- Supervised Learning with scikit-learn – **DataCamp** (2026)
- GitHub Foundations Track – **DataCamp** (2026)
- Network Support and Security Course – **Cisco** (2025)

Publications

- Bilan, E. F., Manduriaga, E. T., et al. (2026). RNNet-MST: A ResNet-50 with Multi-Scale Transformer Blocks for Pulmonary Nodule Classification and Attention-Based Localization on Chest X-Ray Images. *Diagnostics*, 16(10), 1574. DOI: 10.3390/diagnostics16101574
- Undergraduate thesis developed through a PLM–NCKU international research collaboration with the Computer Vision and Multimedia Laboratory at National Cheng Kung University, Taiwan.
 - Published in MDPI Diagnostics, a Q1 peer-reviewed journal, indexed in Web of Science and Scopus.

References

Dr. Hernando Salapare III, *Researcher at CNRS - IS2M, France*
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